

MANAS, Kyrgyzstan

WMO: 382200

Lat: 43.067N

Lon: 74.483E

Elev: 637

StdP: 93.90

Time Zone: 6.00 (E06)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.1	-17.1	-22.2	0.6	-19.7	-19.2	0.7	-16.6	7.5	0.6	6.5	0.8	1.2	180	0.520

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.4	37.0	18.2	35.5	17.6	34.0	17.3	20.5	31.4	19.3	30.7	18.5	29.8	3.7	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.9	13.0	25.2	15.9	12.2	23.5	14.9	11.4	22.1	62.3	31.1	57.7	30.5	54.8	29.6	25.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.6	7.2	6.1	-23.6	39.7	3.8	1.3	-26.3	40.7	-28.6	41.4	-30.7	42.2	-33.5	43.1		
(5)			-23.8	21.1	3.8	1.9	-26.5	22.4	-28.8	23.5	-30.9	24.5	-33.6	25.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	11.8	-4.5	-2.1	6.9	13.5	18.6	23.9	26.3	24.7	19.1	11.8	4.3	-2.4	(6)
(7)		DBStd	11.69	5.87	6.49	5.64	4.60	3.55	3.04	2.79	3.18	3.95	4.77	5.92	5.98	(7)
(8)		HDD10.0	1536	448	339	126	19	0	0	0	0	1	36	184	383	(8)
(9)		HDD18.3	3136	706	572	356	152	41	2	0	2	37	206	421	642	(9)
(10)		CDD10.0	2183	0	1	29	126	268	418	505	455	275	93	14	1	(10)
(11)		CDD18.3	739	0	0	0	8	50	169	247	198	61	5	1	0	(11)
(12)		CDH23.3	9308	0	0	7	161	623	1986	3148	2479	804	91	9	0	(12)
(13)		CDH26.7	4603	0	0	0	42	204	974	1746	1301	314	21	1	0	(13)
(14)	Wind	WSAvg	2.8	2.1	2.3	2.9	3.1	3.3	3.5	3.4	3.2	2.9	2.3	2.1	2.2	(14)
(15)	Precipitation	PrecAvg	439	30	34	44	64	55	33	21	13	16	44	45	36	(15)
(16)		PrecMax	658	68	57	83	146	115	95	67	37	55	113	116	73	(16)
(17)		PrecMin	286	7	9	22	12	13	5	2	0	0	5	4	5	(17)
(18)		PrecStd	98	14	11	17	32	28	24	18	11	16	31	26	15	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.2	16.1	24.1	30.9	33.0	37.8	39.8	38.1	34.8	29.2	24.1	13.9	(19)
(20)			MCWB	4.8	8.1	11.8	16.5	16.7	17.4	18.5	18.0	16.3	14.4	12.3	6.9	(20)
(21)		2%	DB	7.2	11.1	21.1	27.2	30.9	35.2	37.8	36.2	32.8	25.8	18.9	10.0	(21)
(22)			MCWB	3.0	6.0	10.6	14.6	16.0	17.4	18.9	17.6	15.6	12.7	10.1	5.0	(22)
(23)		5%	DB	5.1	8.2	18.9	24.9	29.0	33.8	36.0	35.0	30.1	23.1	15.2	7.2	(23)
(24)			MCWB	2.0	4.3	10.2	13.9	15.7	17.3	18.0	17.0	15.1	11.9	8.3	3.5	(24)
(25)		10%	DB	3.1	5.9	15.9	22.2	27.0	32.0	34.2	33.2	28.1	20.6	12.8	5.0	(25)
(26)			MCWB	0.7	3.1	9.0	12.6	15.2	17.1	17.5	16.9	14.5	11.0	7.1	2.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.7	9.1	13.5	18.3	19.5	21.5	22.8	21.7	18.5	14.9	13.0	7.5	(27)
(28)			MCD B	9.7	15.3	21.3	27.3	28.5	31.8	35.6	33.2	30.5	23.8	21.8	12.1	(28)
(29)		2%	WB	3.7	6.3	11.7	16.1	17.9	19.7	20.7	19.7	16.8	13.6	10.6	5.3	(29)
(30)			MCD B	6.6	10.1	18.6	24.0	26.1	29.9	31.7	31.4	29.0	22.9	17.5	9.1	(30)
(31)		5%	WB	2.1	4.6	10.5	14.9	16.8	18.6	19.4	18.6	15.9	12.7	9.0	3.7	(31)
(32)			MCD B	4.4	7.6	17.0	21.5	25.3	28.9	31.6	30.4	27.4	20.7	14.6	6.9	(32)
(33)		10%	WB	0.9	3.2	9.4	13.8	16.0	17.8	18.5	17.7	14.9	11.7	7.5	2.2	(33)
(34)			MCD B	2.9	5.6	15.3	19.8	24.5	27.9	30.8	29.4	26.2	18.7	11.6	4.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.8	9.5	11.3	12.2	13.3	13.9	14.4	14.8	14.6	13.0	10.5	9.3	(35)
(36)			MCDBR	11.5	12.4	15.8	16.9	16.0	16.3	16.7	16.7	17.2	16.9	15.2	12.4	(36)
(37)			MCWBR	8.0	8.0	7.7	7.0	5.6	4.7	4.9	5.1	5.7	6.9	8.0	8.2	(37)
(38)		5% WB	MCDBR	10.4	11.3	13.8	13.9	14.1	14.3	14.7	15.1	15.6	14.1	13.4	11.1	(38)
(39)			MCWBR	7.3	7.5	7.2	6.4	5.7	5.1	5.3	5.7	6.1	6.3	7.3	7.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.345	0.389	0.451	0.477	0.451	0.451	0.443	0.412	0.396	0.419	0.391	0.340	(40)	
(41)		taud	2.160	2.072	1.954	1.929	2.069	2.109	2.142	2.217	2.202	2.043	2.066	2.157	(41)	
(42)		Ebn at Noon	754	779	775	790	827	826	828	843	826	732	682	717	(42)	
(43)		Edh at Noon	98	128	163	181	161	156	149	134	126	130	108	90	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.03	2.95	3.97	5.04	6.28	6.84	6.86	6.38	5.06	3.18	2.00	1.64	(44)	
(45)		RadStd	0.17	0.25	0.33	0.47	0.45	0.35	0.39	0.32	0.24	0.35	0.21	0.13	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)		N/A	N/A	N/A	N/A	N/A	-0.82	N/A	N/A	N/A	N/A

Nomenclature: See separate page